

(a) Grey Level Transformation:

Grey Level Transformation is used to improve the appearance of image by changing the intensity values of pixels. It helps to enhance contrast, highlight important details and making darker images to visible clearly.

(b) Negative Transformation:

Formula : $S = 255 - r$

r	$S = 255 - r$
50	$255 - 50 = 205$
100	$255 - 100 = 155$
150	$255 - 150 = 105$
200	$255 - 200 = 55$

Interpretation:

→ Dark pixels become brighter

→ Bright pixels become darker.

2. (a) Histogram equalization:

Histogram equalization improves image contrast by spreading intensities of pixels across the intensity available range. It helps to detect the objects, segmentation and to visibility of image.

(b)

Intensity	Frequency	Cumulative Frequency	CDF
0	4	4	0.153
1	6	10	0.384
2	10	20	0.769
3	6	26	1.000

$$N = 4 + 6 + 10 + 6$$

$$= 26$$

$$CDF = \frac{\text{cumulative frequency}}{N}$$

$$\frac{4}{26} = 0.153$$

$$\frac{26}{26} = 1.000$$

$$\frac{10}{26} = 0.384$$

$$\frac{20}{26} = 0.769$$

equalized intensities

It is a 4 level image

(0-3)

$$S = (L-1) \times \text{CDF}$$

$$= (4-1) \times \text{CDF}$$

$$= 3 \times \text{CDF}$$

$$3 \times 0.153 = 0.459$$

$$3 \times 0.384 = 1.152$$

$$3 \times 0.769 = 2.307$$

$$3 \times 1.000 = 3$$

Intensity	CDF	Equalized value.
0	0.153	0
1	0.384	1
2	0.769	2
3	1.000	3

a) 3. Gaussian Filtering:

→ Removes noise smoothly without introducing sharp artifacts.

→ Focuses mainly on centre pixels.

(b) Gaussian kernel:

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

given image:

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Applying gaussian filter

$$\frac{1}{16} \left[(1 \times 10) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (4 \times 40) + (2 \times 20) + (1 \times 10) + (2 \times 20) + (1 \times 10) \right]$$

$$= \frac{1}{16} [10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10]$$

$$= \frac{1}{16} [360] = \frac{360}{16} = 22.5$$

rounded value = 23

$\therefore$ The centre pixel 40 changes to 23

1.(a) edge detection:

edge detection finds the boundaries where pixel intensities changes rapidly helping to detect object and recognize the object.

(b) Sobel mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

given image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Applying Sobel filter:

$$= [(-1 \times 10) + (-2 \times 10) + (-1 \times 10) + 0 + 0 + 0 + (30 \times 1) + (30 \times 2) + (30 \times 1)]$$

$$= [-10 - 20 - 10 + 30 + 60 + 30]$$

$$= [-40 + 120]$$

$$= 80$$

$$\boxed{\text{Gradient} = 80}$$

$\therefore$ The value is 80 $\rightarrow$ edge is present.

5. Contrast Stretching:

(a) Contrast Stretching expands the range of pixel intensities as image can help to make darker images as darker and brighter images as brighter.

(b) Formula:

$$S = \frac{r - r_{\min}}{(r_{\max} - r_{\min})} \times 255$$

given $r_{\max} = 150$

$$r_{\min} = 50$$

$$r_{\max} - r_{\min} = 150 - 50 = 100$$

For $x = 50$

$$S = \frac{50 - 50}{100} \times 255 = 0 \times 255 = 0$$

For $x = 100$

$$S = \frac{100 - 50}{100} = \frac{50}{100} = 0.5 \times 255 = 127.5 = 128$$

For $x = 150$

$$S = \frac{150 - 50}{100} = \frac{100}{100} = 1 \times 255 = 255$$

x	Transformed values
50	$0 \times 255 = 0$
100	$0.5 \times 255 = 128$
150	$1 \times 255 = 255$

$\therefore$ Contrast stretching changes range
[50-150] to [0 to 255]